

Introduction to Computational Thinking and Data Science (ICTDS)

Academic Year: 2025/2026

Milestone 3: Climate Stability Analysis: Bragança vs Viseu

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Group: 11

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1 Introduction

Goal: A cold-storage logistics operator requires a site selection decision between two Portuguese cities — Bragança and Viseu — based on climate suitability. The core objective of this analysis is to provide a data-driven recommendation on which city offers the cooler, more stable, and operationally predictable thermal environment.

Cold-storage operations are highly sensitive to ambient temperature: warmer and more variable climates translate directly into higher compressor loads, elevated energy consumption, and greater risk of temperature excursions. To solve this, we analysed daily meteorological records from 2018 to 2025 (2,922 observations per city) sourced from Open-Meteo. This report transitions from earlier exploratory descriptive statistics to rigorous inferential statistical validation. We seek to determine whether the observed thermal differences are statistically robust and to quantify their operational impact for a continuous cold-chain facility.

2 Dataset Description

The meteorological data represents gridded reanalysis model output retrieved from the Open-Meteo historical weather API, ensuring no missing values over the requested study period.

2.1 Variables

The following features were available or derived from the raw data for both cities:

Table 1: Dataset Variables

Variable	Type	Description
temperature_2m_max	Numerical	Daily maximum temperature at 2m (°C)
temperature_2m_min	Numerical	Daily minimum temperature at 2m (°C)
temperature_2m_mean	Numerical	Daily mean temperature at 2m (°C)
precipitation_sum	Numerical	Total daily precipitation (mm)
wind_speed_10m_max	Numerical	Max wind speed at 10m (km/h)
wind_gusts_10m_max	Numerical	Max wind gusts at 10m (km/h)
shortwave_radiation_sum	Numerical	Daily solar radiation sum (MJ/m ²)
sunshine_duration_hrs	Derived	Sunshine duration (hours; converted from seconds)
relative_humidity_2m_mean	Numerical	Mean relative humidity at 2m (%)
cloudcover_mean	Numerical	Mean cloud cover (%)
day_length_hrs	Derived	Length of day in hours (sunset – sunrise)
DTR	Derived	Diurnal Temperature Range = max – min (°C)

2.2 Basic Summary

The datasets span from 2018-01-01 to 2025-12-31. After removing redundant time columns and excluding the incomplete year of 2026 to avoid seasonal bias, the dataset contains exactly **2,922 observations per city** across 12 variables.

3 Methodology

3.1 Key Questions and Success Criteria

In accordance with the transition to formal validation in Milestone 3, our analysis is guided by the following key questions:

- **Statistical Significance:** To what extent are the observed differences in mean seasonal temperatures and thermal variability between Bragança and Viseu statistically significant, or are they attributable to sampling variation?
- **Robustness and Uncertainty:** How robust are our initial site interpretations when subjected to statistical hypothesis testing and interval estimation?
- **Operational Meaningfulness:** Beyond statistical significance, are the observed temperature differentials practically meaningful for refrigeration demand, energy consumption, and thermal shock risk in a logistics environment?
- **Controlled Variation:** Does controlling for seasonal and monthly fluctuations alter the perceived thermal advantage of one location over the other?
- **Operational Predictability:** How stable is the thermal profile across different study periods, and how does this affect long-term operational cost efficiency projections?

Success is defined by delivering a statistically justified, evidence-based strategic recommendation that quantifies the uncertainty and operational trade-offs of selecting Bragança versus Viseu.

3.2 Analysis approach

Our methodology followed a layered structure, transitioning from exploratory data analysis to inferential statistical validation:

1. **Preprocessing:** Converting time metrics, deriving day length, and assigning categorical seasons (Winter, Spring, Summer, Autumn) to prevent conflating opposite temperature regimes.
2. **Exploratory Data Analysis (EDA):** Evaluating distribution shapes, identifying feature drivers via Pearson correlations, and profiling thermal envelopes and positive deviations.
3. **Hypothesis Testing (Normality & Significance):** Applying the Shapiro-Wilk test for normality and the Mann-Whitney U test to check if inter-city temperature differences are statistically significant ($p < 0.05$).
4. **Uncertainty & Effect Size:** Utilizing 95% Confidence Intervals and Cohen’s d to quantify the practical magnitude of the temperature differences between cities.
5. **Variability & Risk:** Applying Levene’s Test (F-Distribution) to assess which city experiences significantly higher temperature volatility.
6. **Categorical Analysis:** Constructing a contingency table and performing a Chi-Square test to evaluate the disproportionate occurrence of extreme heat days ($> 30^{\circ}\text{C}$).
7. **Regression Modeling:** Deploying an Ordinary Least Squares (OLS) Multiple Linear Regression model with dummy variables to control for seasonality, isolating the exact coefficient of shortwave radiation on ambient temperature.

4 Results

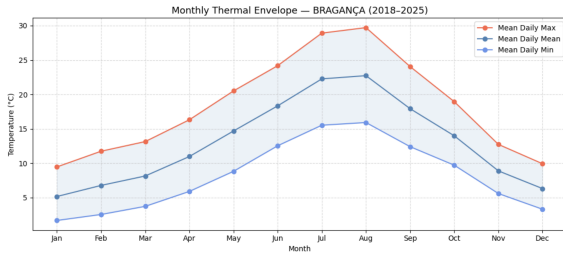
4.1 Data Preprocessing and Exploratory Data Analysis (EDA)

Before extracting key statistics, the raw data underwent structured preprocessing to ensure operational relevance. Redundant `time` columns were removed, and datetime conversions allowed the extraction of `year` and `month`. `sunshine_duration` was converted to hours, and `day_length_hrs` was derived from sunrise/sunset timestamps. To prevent conflating opposite thermal regimes, data was binned into four astronomical seasons, and the incomplete year of 2026 was excluded.

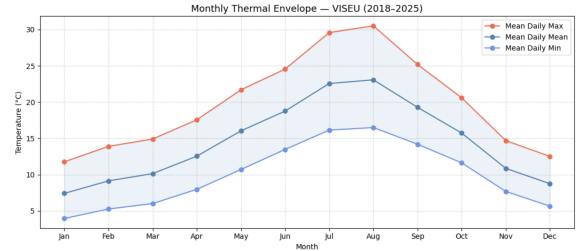
Exploratory Data Analysis (EDA) revealed several foundational insights:

- **Thermal Envelope & Distributions:** Bragança’s Winter interquartile range (IQR) spans roughly 3.5°C – 9.0°C , while Viseu’s sits approximately 2°C higher. In Summer, the IQRs overlap substantially near 21°C . Viseu’s thermal envelope is uniformly elevated, with winter minimums consistently 2 – 3°C warmer than Bragança’s.

- **Diurnal Temperature Range (DTR):** The average daily swing between maximum and minimum temperatures is similar between cities, with the largest gap occurring in Spring (Bragança’s DTR is 0.67°C wider).
- **Feature Drivers:** Pearson correlation heatmaps established that shortwave radiation ($r \approx +0.80$) and relative humidity ($r \approx -0.78$) are the strongest symmetric drivers of DTR. Day length strongly predicts the Stability Index ($r = -0.72$ at Viseu, $r = -0.55$ at Bragança). Wind speed and gusts are largely uncorrelated with thermal load ($|r| \leq 0.31$).
- **Extreme Events & Positive Deviations:** A seasonal IQR-based anomaly detection flagged 7 extreme hot-and-dry days for Bragança (heavily front-loaded into Spring) and 6 for Viseu (spread across Spring, Summer, and notably Autumn). Positive deviation analysis showed Viseu experiences larger unplanned Summer temperature spikes (up to 12.3°C above the monthly mean maximum, compared to Bragança’s $\approx 10.5^\circ\text{C}$).

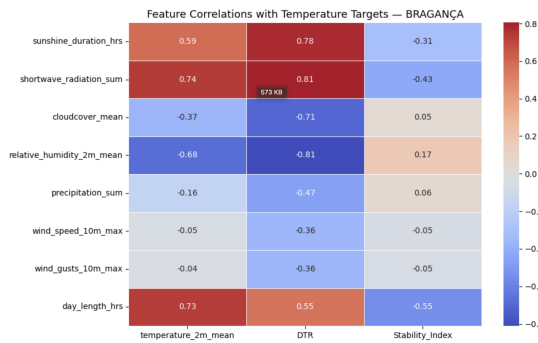


(a) Thermal Envelope: Bragança

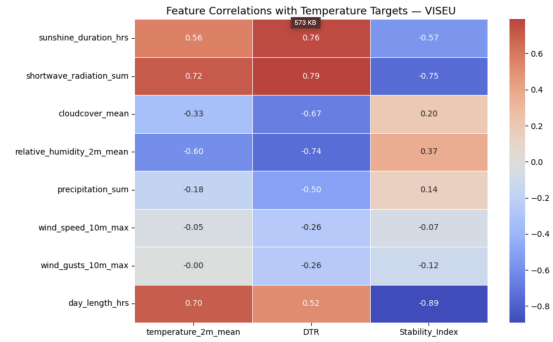


(b) Thermal Envelope: Viseu

Figure 1: Monthly mean daily max, mean, and min temperatures. Viseu’s thermal envelope is uniformly elevated relative to Bragança.



(a) Feature Correlations: Bragança



(b) Feature Correlations: Viseu

Figure 2: Pearson heatmaps show shortwave radiation and day length as massive drivers of temperature and instability, whereas wind is negligible.

4.2 Key Statistics

Table 2 confirms the EDA observations: Bragança is cooler than Viseu in every season without exception. The gap is largest in Winter (2.34°C) and Autumn (1.69°C), and smallest in Summer (0.35°C).

Table 2: Seasonal Mean Daily Temperature & DTR ($^{\circ}\text{C}$) — Bragança vs Viseu (2018–2025)

City	Winter		Spring		Summer		Autumn	
	Mean	DTR	Mean	DTR	Mean	DTR	Mean	DTR
Bragança	6.07	7.82	11.28	10.50	21.14	12.95	13.61	9.33
Viseu	8.41	7.75	12.91	9.83	21.49	12.87	15.30	8.99
<i>Diff (Br–Vi)</i>	–2.34 $^{\circ}\text{C}$		–1.63 $^{\circ}\text{C}$		–0.35 $^{\circ}\text{C}$		–1.69 $^{\circ}\text{C}$	

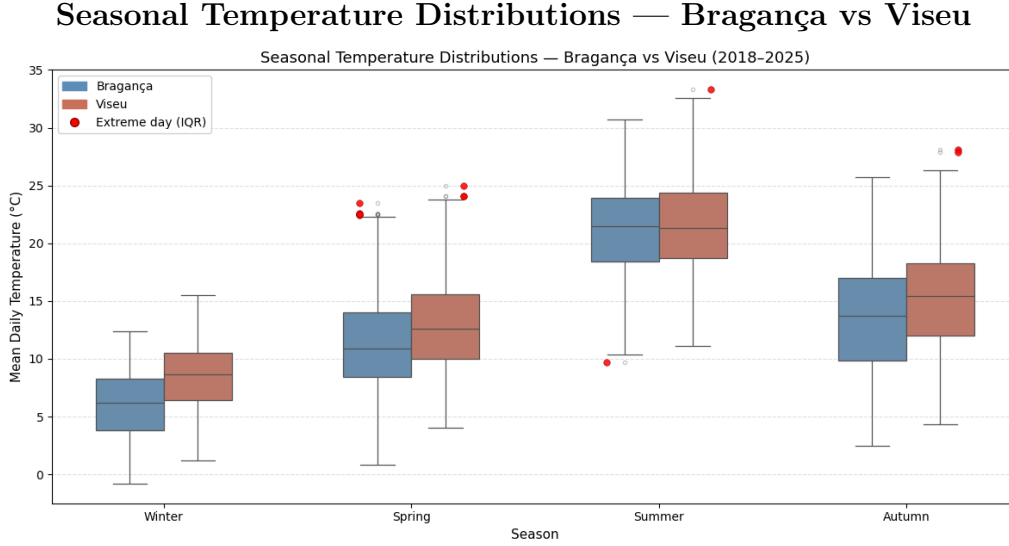


Figure 3: Side-by-side seasonal boxplots. Bragança (left, blue) is cooler in every season; the gap is widest in Winter and Autumn, narrowing sharply in Summer. Red points mark IQR-flagged extreme events.

A composite **Temperature Stability Index** (combining within-season SD, mean DTR, and inter-annual CoV) revealed that while Viseu is more stable in Winter and Autumn, Bragança holds the stability advantage in Summer (0.238 vs Viseu’s 0.216) — the operationally critical peak load season.

4.3 Inferential Statistical Validation

4.3.1 Normality & Hypothesis Testing

A Shapiro-Wilk test confirmed that temperature distributions in both cities are strictly non-normal across all seasons. Consequently, a Mann-Whitney U test was applied to test the significance of the observed mean differences.

Table 3: Hypothesis Testing: Mann-Whitney U Results

Season	p-value	Significant ($p < 0.05$)	Conclusion
Winter	7.45×10^{-46}	True	Structurally Different
Spring	4.52×10^{-13}	True	Structurally Different
Summer	0.236	False	Not Significant
Autumn	6.74×10^{-11}	True	Structurally Different

The differences in temperatures are statistically significant for all seasons *except* Summer. This implies refrigeration demands will be structurally different depending on the city for the majority of the year, while summer demands will be similar.

4.3.2 Uncertainty & Effect Size

To understand the practical scale of these differences, Cohen’s d and 95% Confidence Intervals were calculated.

Table 4: Seasonal Temperature: Effect Size & Uncertainty

Season	Mean Diff (°C)	95% CI [Lower, Upper]	Cohen’s d	Effect Size
Winter	−2.34	[−2.63, −2.05]	−0.84	Large
Spring	−1.63	[−2.05, −1.21]	−0.40	Small
Summer	−0.35	[−0.76, 0.06]	−0.09	Negligible
Autumn	−1.69	[−2.16, −1.21]	−0.37	Small

The Summer confidence interval confirms that at worst, the peak temperature difference is only 0.76°C. The negligible Cohen’s d (−0.09) proves summer compressor loads will safely mirror each other. Conversely, Winter displays a **Large** effect size, confirming a massive practical reduction in baseline thermal load for Bragança.

4.3.3 Variability & Categorical Risk

Levene’s test was utilized to quantify temperature volatility via variance. Bragança showed a slightly higher variance exclusively in Winter ($p = 0.03$). However, the categorical analysis on Extreme Heat Days proves Viseu is significantly more dangerous during peak stress events.

Table 5: Contingency Table & Chi-Square Test: Extreme Heat Frequency

City	Extreme Days (> 30°C)	Normal Days ($\leq$ 30°C)
Bragança	277	2645
Viseu	343	2579

The Chi-Square test yielded a statistic of 7.62 with a p -value of 5.76×10^{-3} . This proves that Viseu’s disproportionate occurrence of heat waves (roughly 19% more than Bragança) is statistically meaningful, rather than a fluke of the sample data.

Seasonal Temperature Distribution with Extreme Events — Viseu

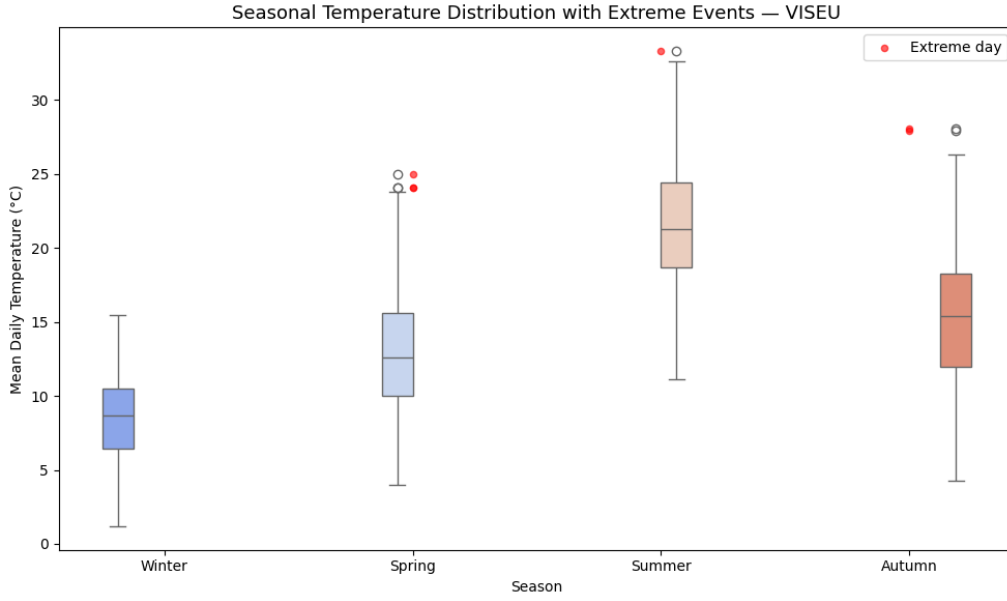


Figure 4: Seasonal boxplots for Viseu. Unlike Bragança (whose extremes are constrained to Spring), Viseu distributes IQR-flagged extremes across three seasons, introducing a longer operational exposure window to anomalous heat.

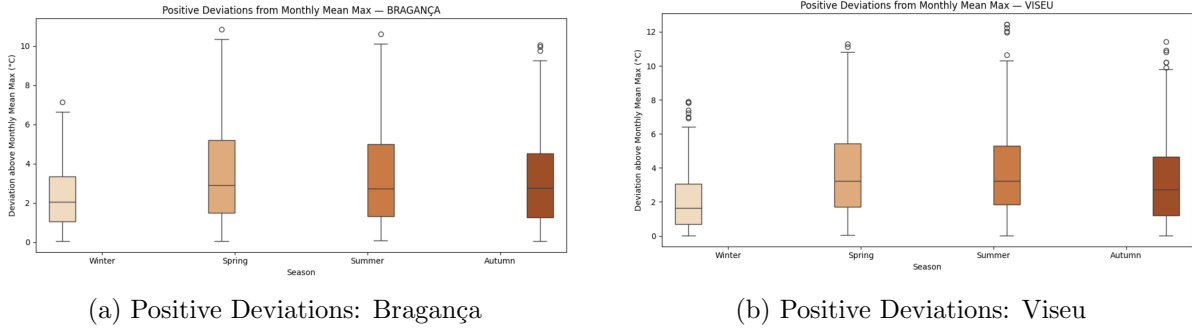


Figure 5: Positive deviations from the monthly mean maximum. Viseu shows larger deviation spreads in Winter and Summer (outliers reaching 12.3°C above average, compared to Bragança's 10.5°C).

4.3.4 Multiple Regression: Controlling for Seasonality

To accurately forecast cooling requirements, an OLS model with seasonal dummy variables isolated the effect of shortwave radiation from general seasonal shifts.

Table 6: OLS Regression: Shortwave Radiation & Seasonality vs Temperature

Parameter	Bragança	Viseu
Adjusted R^2	0.734	0.696
Base Intercept (Winter Baseline)	3.3315°C	5.5738°C
Radiation Coefficient (Slope)	0.3684	0.3601
Season (Spring) Shift	+1.0481°C ($p < 0.01$)	+0.6439°C ($p < 0.01$)
Season (Summer) Shift	+8.3544°C ($p < 0.01$)	+6.7839°C ($p < 0.01$)
Season (Autumn) Shift	+5.8359°C ($p < 0.01$)	+5.3549°C ($p < 0.01$)

5 Discussion

5.1 Operational Implications

The quantitative findings translate to concrete operational consequences for a mixed cold-storage logistics facility:

- **Winter:** Bragança’s statistically verified cooler baseline (intercept of 3.33°C vs Viseu’s 5.57°C) creates occasional freeze-risk exposure rather than heat exposure. Here, Viseu holds a marginal advantage by reducing passive heat gain, but Bragança saves massive baseline energy.
- **Summer:** Summer is the most critical season. Bragança holds a clear advantage: the mathematical models prove peak loads will be practically identical (Cohen’s $d = -0.09$), yet Bragança scores a higher Stability Index (0.238 vs 0.216) indicating more predictable day-to-day conditions.
- **Autumn/Spring:** Viseu records extreme heat events late into Autumn, introducing a longer operational exposure window to anomalous heat when cold-storage demand should normally begin to ease.

5.2 Variance, Volatility, and the Solar Synergy

Levene’s test proved higher variance translates directly to operational risk. Predictability is paramount. By including radiation and seasonality, our regression model achieved an Adjusted R^2 of 0.734 for Bragança. This means 73% of Bragança’s ambient temperature is highly predictable.

Because we isolated the specific Shortwave Radiation coefficient (0.3684°C per unit for Bragança), we can accurately forecast the thermal offset a rooftop solar array will provide. By physically shielding the roof with panels, the reduced radiation can be multiplied by this 0.3684 slope to calculate the precise drop in ambient compressor load, creating a defensively modeled, highly optimized thermodynamic loop.

5.3 Limitations

While robust, this analysis relies on historical data (2018–2025) and cannot account for macro-climate change trends. Unmodeled environmental factors (Omitted Variable Bias)—such as

local wind patterns or urban heat island effects—account for the remaining 26% of unexplained temperature variance. Furthermore, the solar thermal shielding assumes panels will not trap heat, requiring a dedicated engineering validation.

6 Conclusion

Based on comprehensive exploratory analysis and rigorous inferential statistical validation, **Bragança is definitively the superior strategic choice for the cold-storage logistics hub.**

While Viseu and Bragança experience nearly identical peak cooling loads during the summer (statistically validated as a negligible difference), Bragança consistently provides a massively colder baseline during the rest of the year. Crucially, the Chi-Square tests proved Bragança suffers roughly 19% fewer extreme heat events ($>30^{\circ}\text{C}$). For a continuous cold-chain operation, this translates to significantly reduced CAPEX (smaller peak-load HVAC and generator requirements) and heavily optimized OPEX over the facility’s lifespan.